Collaborator or Assistant?

How AI Coding Agents Partition Work Across Pull Request Lifecycles

Agent-associated PRs now enter real projects at scale.
What does the visible trace show when they merge?

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Our contribution: a map of where to look

Raw PR metadata records visible actors and events - not hidden authorship, authorization, or accountability. We turn it into a map.

1 **Each PR is a lifecycle**

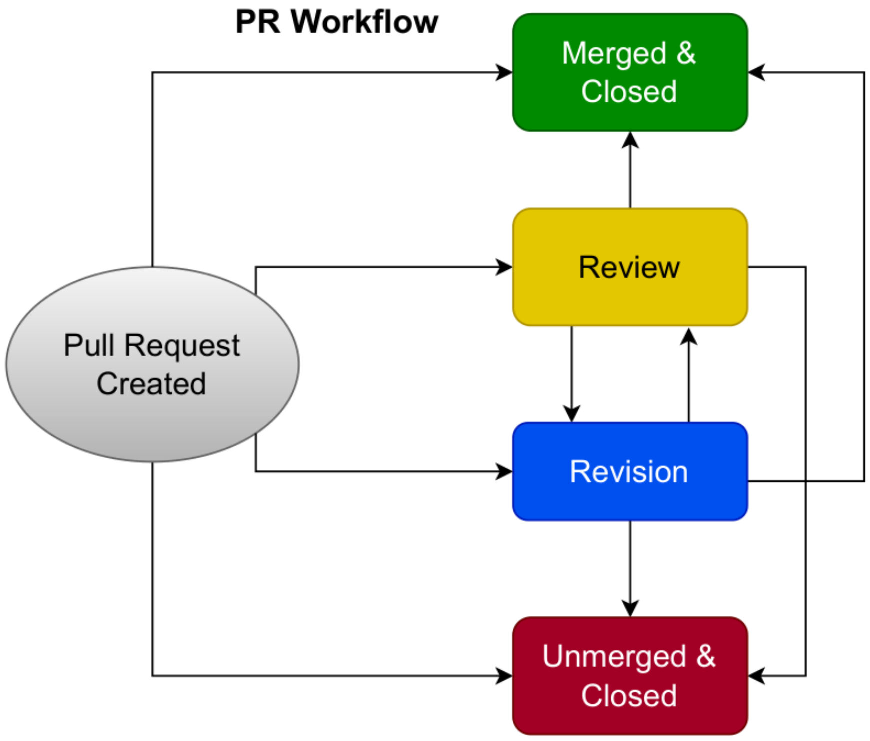
A process to characterize — not a row in a table.

2 **Timeline maps the pathways**

State-machine transitions recover the routes a PR can take.

3 **Three separable observables**

Visible attribution - review trace - merge execution: what surfaces where.



The paper's PR-lifecycle state machine (the map).

Same data, a different lens

-same PRs, same metadata
-timeline vs. review dataset

Same metadata as AIDev (Li et al., 2025 & 2026), but not the same construct: the commit timeline vs. the human-approved review record.

Tool	Agent visibility in first-commit trace	Recorded human-approved review trace
Copilot	99.2%	66.4%
Devin	98.4%	44.7%
Cursor	96.1%	28.1%
Claude	*AIDev: 100%* 4.1%*	19.4%
OpenAI Codex	*AIDev: 100%* 0.1%	3.9%

**Claude is the fragile case: n=338 and 57% of labels flip under attribution-aware re-detection. Codex carries the obscured-agency pattern at scale (n=20,857).*

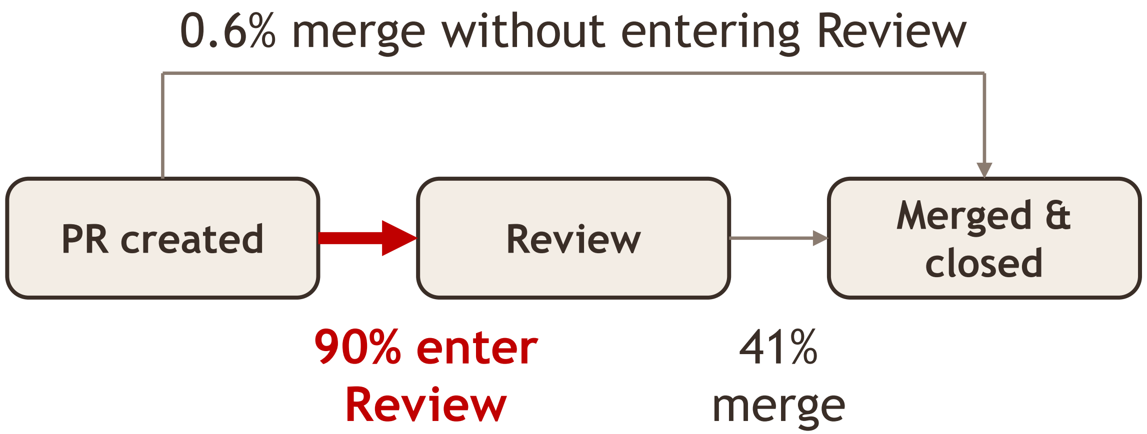
The gap between agent association and human-surfaced attribution is the governance phenomenon.

The twist: merge stays visible; Review is often never entered

Same 29,585 PRs, routed through the lifecycle map — 89.6% lack a recorded human-approved review. Here is where the review disappears.

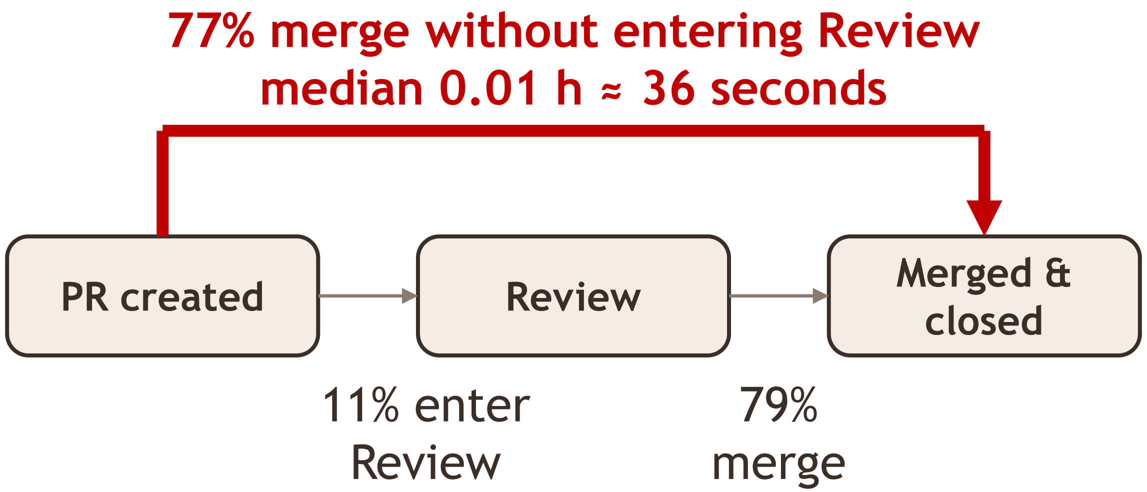
Copilot

Collaborator end of the spectrum



OpenAI Codex

Assistant end of the spectrum



The 17× spread is topology: Copilot's road runs through Review; Codex's bypasses it.

Edge % = P(next phase | current phase); unmerged paths omitted. Route ≠ diligence: off-record scrutiny leaves no trace.

Collaborator or Assistant? See the poster

Three separable questions: who gets credit · where recorded review appears · who executes the merge.

What should platforms, tools, and repositories log before AI-assisted code becomes software people depend on?

- Merge ≠ decision — 54 merged PRs (0.24%) are automation-authorized; the trace records the mechanism.
- The construct correction — human-surfaced trace, not hidden authorship.
- The accountability chain — trace → actor → accountable decision · code & data at the QR.

Come to the poster — the instrument, and what it reveals.

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